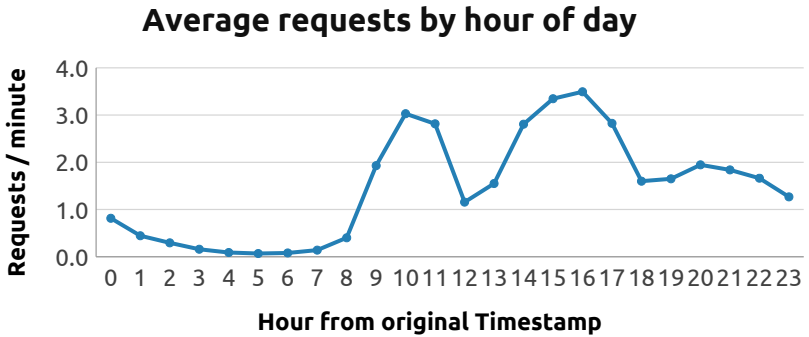
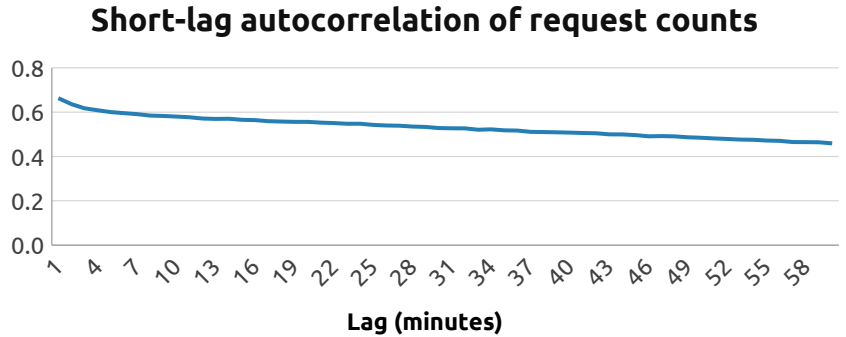
EDA of the dataset: BurstGPT

Exploratory Data Analysis (1): Demand over time

233,617 conversation requests (~110 days). Forecast every minute for the next 10 minutes.





- **Short-term dependence: request ACF is 0.663 at 1 minute and 0.580 at 10 minutes.**
- **Model A uses recent demand history and time of day.**
  1/2/3-minute lags, 5/15/30/60-minute totals, and recent demand variability.

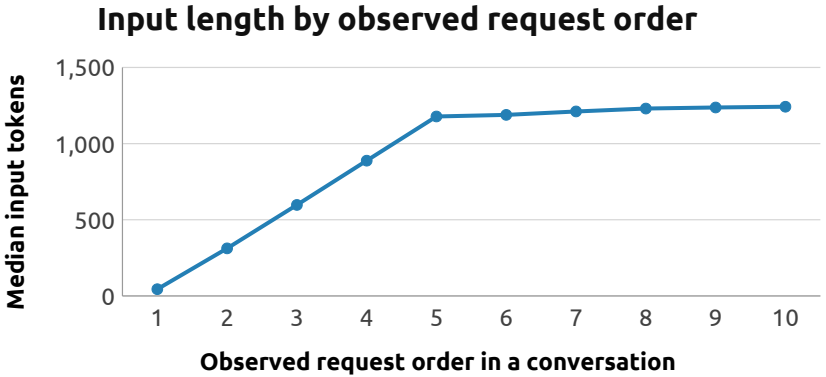
  10-min input/output totals are zero at 15.81% of eligible times
  (5-min target: 22.33%, same 158,328 forecast times).

Source: 08_hour_of_day_profile.csv, 08_autocorrelation_all_lags.csv and 08_forecast_horizon_summary.csv.

EDA of the dataset: BurstGPT

Exploratory Data Analysis (2): Conversation continuity

Conversation evidence	Value
Unique conversation IDs	55,920
Conversations with 2+ requests	64.53%
Requests in repeated conversations	91.51%
Observed follow-up gap, median	131 sec



- **Conversation state is a plausible source of additional information.**
  Observed follow-up gaps within 10 min: 76.69% (within 5 min: 66.10%).

- **Models B/C add conversation information to the same aggregate baseline.**
  B: active counts over 5/15/30 min. C: recency, request count and input-length state.

Order groups contain different conversations. Follow-up gaps describe observed pairs, not return probabilities.

Source: summary.json, 06_input_length_by_observed_turn.csv and 06_followup_gap_thresholds.csv.

EDA of the dataset: BurstGPT

Exploratory Data Analysis (3): Testing added value

Every minute: next-10-min input and final-output totals for arrivals in [t, t+600 s).

Exploratory Spearman correlations

Past predictor	10m input	10m output
5-min request count	0.822	0.832
15-min active conversations	0.847	0.866

Past request count vs. active conversations (5 min)
Spearman correlation = 0.978

Pairwise correlation does not establish added value.

Controlled feature comparison

Model	Feature set
A	Aggregate demand history
B	A + active conversation counts
C	B + conversation state summaries

Same XGBoost family, 60/20/20 split and tuning budget.
Evaluate input/output separately with MAE and RMSE.

- **13.90% of 10-min output targets include responses completing at/after window end.**
  Require a closed 10-min window and completed responses before using training labels.

Source: 08_spearman_correlations.csv and summary.json. Full-period EDA informed this design. Model gains remain untested.